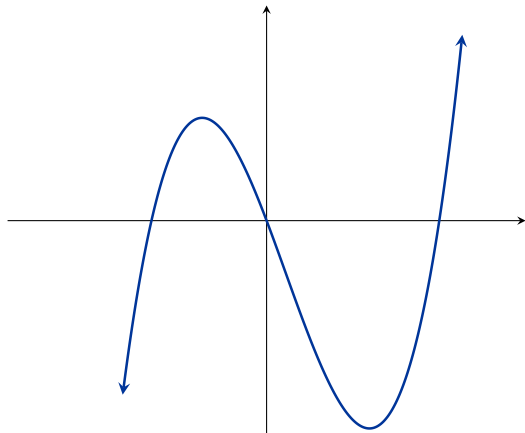


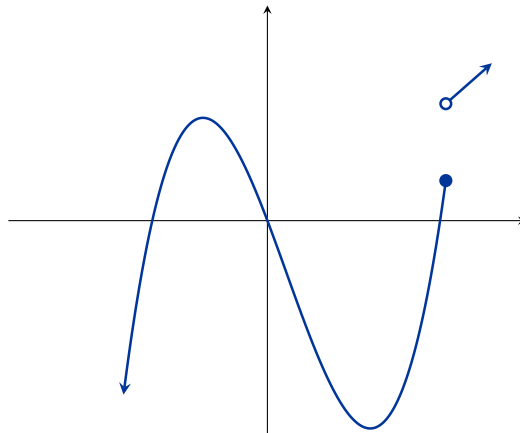
3.4: Graphs of Polynomial Functions

Graphs of polynomials are smooth and continuous with infinite end behavior.

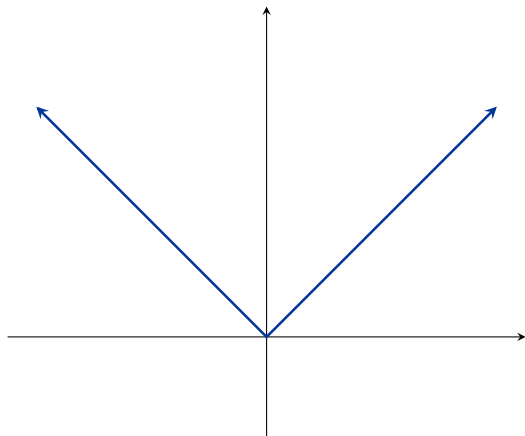
Smooth and continuous



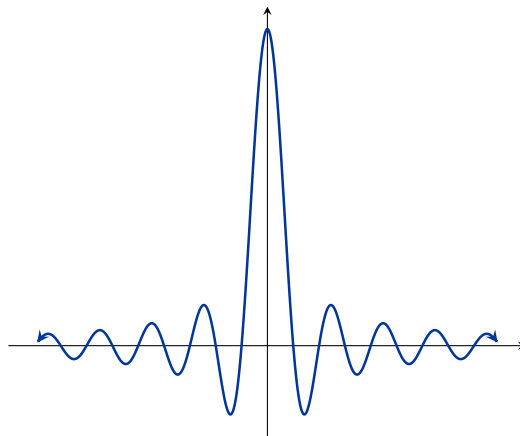
Not continuous



Not smooth



Finite end behavior



Example. Find the x - and y -intercepts of the following functions:

$$f(x) = 2(x-4)(x+1)(x-6)$$

x -intercept solve $f(x) = 0$

$$x-4=0 \Rightarrow x=4 \rightarrow \boxed{(4,0)}$$

$$x+1=0 \Rightarrow x=-1 \rightarrow \boxed{(-1,0)}$$

$$x-6=0 \Rightarrow x=6 \rightarrow \boxed{(6,0)}$$

y -intercept Evaluate $f(0)$

$$f(0) = 2(0-4)(0+1)(0-6)$$

$$= 2(-4)(1)(-6)$$

$$= 48 \rightarrow \boxed{(0,48)}$$

$$h(x) = x^4 - x^2$$

x -intercept solve $h(x) = 0$

$$x^4 - x^2 = x^2(x^2 - 1) = 0$$

$$x^2 = 0 \rightarrow x = 0 \rightarrow \boxed{(0,0)}$$

$$x^2 - 1 = 0 \rightarrow x = \pm 1 \rightarrow \begin{matrix} \boxed{(-1,0)} \\ \boxed{(1,0)} \end{matrix}$$

y -intercept Evaluate $h(0)$

$$h(0) = 0^4 - 0^2 = 0 \rightarrow \boxed{(0,0)}$$

$$g(x) = 4x(x-2)^2(x+1)$$

x -intercept solve $f(x) = 0$

$$4x = 0 \Rightarrow x = 0 \rightarrow \boxed{(0,0)}$$

$$(x-2)^2 = 0 \Rightarrow x = 2 \rightarrow \boxed{(2,0)}$$

$$x+1 = 0 \Rightarrow x = -1 \rightarrow \boxed{(-1,0)}$$

y -intercept Evaluate $g(0)$

$$g(0) = 4(0)(0-2)^2(0+1)$$

$$= 0 \rightarrow \boxed{(0,0)}$$

$$j(x) = x^3 + 2x^2 - 9x - 18$$

$$= x^2(x+2) - 9(x+2)$$

$$= (x^2 - 9)(x+2)$$

x -intercept solve $j(x) = 0$

$$x^2 - 9 = 0 \rightarrow x = \pm 3 \rightarrow \begin{matrix} \boxed{(-3,0)} \\ \boxed{(3,0)} \end{matrix}$$

$$x+2 = 0 \rightarrow x = -2 \rightarrow \boxed{(-2,0)}$$

y -intercept Evaluate $j(0)$

$$j(0) = 0^3 + 2(0)^2 - 9(0) - 18$$

$$= -18 \rightarrow \boxed{(0,-18)}$$

Definition. (Multiplicity of a Root)

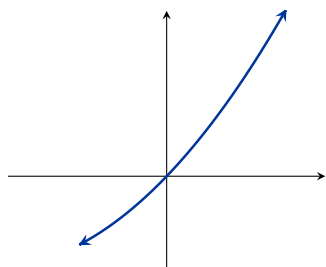
If a polynomial contains a factor of the form $(x - h)^p$, then the behavior near the x -intercept h is determined by the power p . We say that $x = h$ is a zero of **multiplicity** p .

When p is,

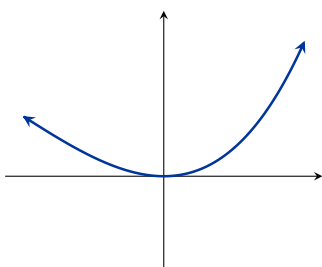
- even, the graph will touch the x -axis
- odd, the graph will cross the x -axis

The degree of the polynomial is the sum of the multiplicities.

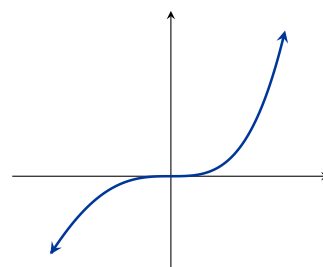
Single zero



Zero with multiplicity 2



Zero with multiplicity 3



Example. For the following, find the zeros, and the multiplicities of each:

$$f(x) = (x + 2)^3(x - 3)^2 \underbrace{(x^2 - 1)}_{(x-1)(x+1)}$$

$$x = -2, \text{ multiplicity } 3$$

$$x = 3, \text{ multiplicity } 2$$

$$x = 1, \text{ multiplicity } 1$$

$$x = -1, \text{ multiplicity } 1$$

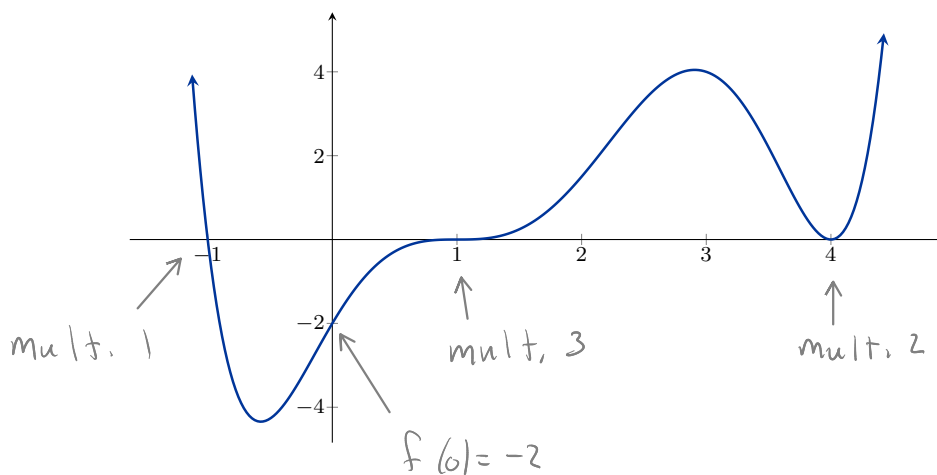
$$g(x) = x^5(x - 1)^3(x + 2)$$

$$x = 0, \text{ multiplicity } 5$$

$$x = 1, \text{ multiplicity } 3$$

$$x = -2, \text{ multiplicity } 1$$

Example. Find a formula for the polynomial of minimum degree represented in the graph below:



$$f(x) = a(x+1)(x-1)^3(x-4)^2 \quad \text{degree 6}$$

$$\begin{aligned} f(0): \quad -2 &= a(0+1)(0-1)^3(0-4)^2 \\ -2 &= a(1)(-1)(16) \quad \Rightarrow \quad a = \frac{-2}{-16} = \frac{1}{8} \end{aligned}$$

$$\Rightarrow f(x) = \frac{(x+1)(x-1)^3(x-4)^2}{8}$$